

LAB ASSIGNMENT # 01

Introduction to Flowcharts and Java Programming



CSE110 | Programming Language I

	LAB TASKS	HOME TASKS
CODING	06	05
FLOWCHART	04	05

NOTE: You need to submit only the Home Tasks. Submit all the Flowchart or Tracing tasks hand drawn or handwritten respectively to your Lab Instructors before the next lab. Submit all the Homework Coding Tasks in the Google Form shared on buX.

Lab Policy: [Lab Policy Student Version - Summer 2025 Onward](#)
Must Submit the Lab Agreement Form: [Form Link](#)

LAB TASKS

[NO NEED TO SUBMIT]

FLOWCHART TASKS

Question: 1

Design a flowchart to take a number as input, store it in a variable, and print its cube.

Question: 2

Create a flowchart to convert a temperature in Fahrenheit to Celsius. The user should input the temperature in Fahrenheit, and the flowchart should display the corresponding temperature in Celsius. The formula to convert Fahrenheit to Celsius is: $C = (F - 32) \times 5/9$

Question: 3

Jamal is struggling with his math homework, which requires him to calculate the volume of water needed to fill a cylinder completely. Unable to solve the problem on his own, he turns to you, his friend who excels in mathematics and computer science. To assist him, you decide to create a flowchart that takes the cylinder's height (H), radius (R), and the current height of water (h) as user inputs and determines the additional volume of water needed to fill the cylinder to its full capacity.

$$[\text{Volume of Cylinder} = \pi * R^2 * H]$$

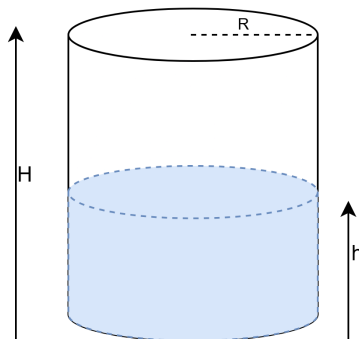


Fig. 1

Question: 4

Create a flowchart to find the average of three numbers. The flowchart should take three inputs from the user and display their average.

For this course, we'll be using **DrJava** as IDE for Java Coding:

[Link to JDK and DrJava](#)

DrJava Installation Guide:

[Windows Installation](#)

[MAC Installation](#)

CODING TASKS

Question: 1

Find out which of the following are legal identifiers in Java, and which are not. Also, explain why the invalid ones are invalid. You can try to define each of these (items a to l below) as a variable in the Dr java interaction pane and find out.

The first one is done for you as an Example. If you want to define hungry, you have to try

int hungry;

a) hungry	b) 2AB	c) 312.2	d) MOBILE
e) "Ans"	f) \$30	g) Yes/No	h) student-id
i) A+3	j) 'X'	k) public	l) student_id

Question: 2

Write the Java code for the following:

- 2.1 Declare an **integer** variable. Initialize it with some value of your choice and print it to check the value has been stored properly.
- 2.2 Declare and initialize another **integer** variable. Add this to the first one and print out the result. Verify that the addition has been done correctly.
- 2.3 Now print the product and division of the two **integer** numbers.
- 2.4 Repeat exercises 2.1, 2.2, and 2.3 for variables of data type **double**. Verify your answers.

2.5 Repeat exercises 2.1, 2.2, and 2.3 for one **double** data type and one **integer** datatype. Verify your answers.

2.6 Repeat exercises 2.1 and 2.2 for variables of data type **String**. How does the addition operator work for Strings? What if the first variable is an integer and the second is a String and vice versa?

Question: 3

Write a Java code that calculates and prints the circumference and area of a circle with a radius of 4 units. [Hint: use Math.PI to get the value of π]

Question: 4

Write a Java code where given an integer, we need to print the last 2 digits of that number.

Question: 5

A local tailor named Jamal often receives measurement details from his foreign customers in inches, but his measuring tools only use meters. To make his work easier, Jamal decides to ask his programmer friend (you) to write a simple Java program that will convert inches into meters.

[NOTE: 1 inch = 0.0254 meters]

Test Data:

Given a value for inch: 1000

Expected Output:

1000 inch is 25.4 meters

Question: 6

Write a Java program declaring two integer variables and initializing them. Your task is to swap the values of these two variables. You must complete it using two different approaches.

- a. By creating a third variable.
- b. Without creating any other variables.

Home TASKS

FLOWCHART TASKS

Question: 1

Design a flowchart to take an integer consisting of 4 digits as input and print the last 2 digits of that number.

Question: 2

Design a flowchart to take an integer consisting of 4 digits as input and print the first 2 digits of that number.

Question: 3

Design a flowchart to take three integers a , n , d as input where a is the first term of an arithmetic sequence, n is the number of terms and d is the common difference between the terms in the sequence, and print the n -th term of the arithmetic sequence.

Question: 4

Design a flowchart to take two numbers a and c as input (look at the image below), and print the area of the portion colored in green. In the following image, a is the radius of the circle, and c is the length of the sides of the square.

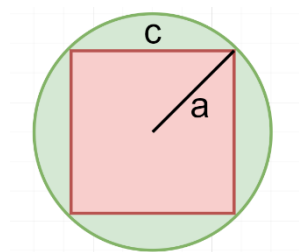


Fig. 2

Question: 5

Design a flowchart to take a number a as input (look at the image below), and print the area of the portion colored in red. In the following image, a is the diameter of the circle.

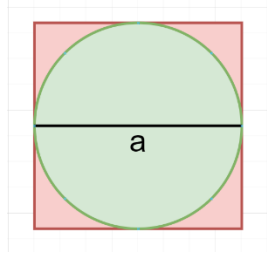


Fig. 3

CODING TASKS

Question: 1

NASA has recently shared the total time (in minutes) since one of its space missions was launched. As a software engineer working at the mission control center, your task is to help them calculate how many years and days have passed since the launch. Therefore, write a Java program that can do the conversion.

NOTE: For simplicity, assume each year has exactly **365 days**.

Test Data:

Given the number of minutes: 3456789

Expected Output:

3456789 minutes is approximately 6 years and 210 days

Question: 2

Suppose, you have three integer variables: a, b, c. Your first task is to assign the values 2, 5, 8 in these three variables. Next, you need to calculate and display the value of variable d using the following formula:

$$d = (2b \times \frac{c-a}{3}) + 7$$

Write a Java program based on this mentioned scenario that prints the value of d after calculation. **[Answer: 27]**

Question: 3

Rafi has just started learning Java programming. His teacher gives him a simple challenge to test his understanding of integer manipulation and modulus operations.

The task is to display the two rightmost digits of his student ID in reverse order.

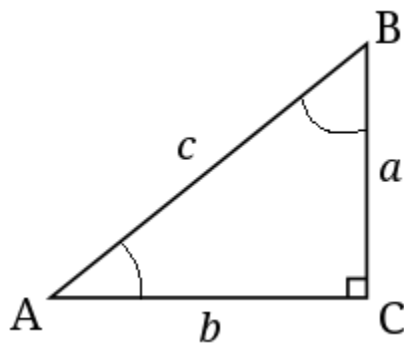
Rafi knows his student ID is **1000054943**, and he must write a Java program that prints the digits **3** and then **4**, since those are the two rightmost digits reversed.

Question: 4

Rafi is helping his friend place a Wi-Fi router in a rectangular room. He wants to know the shortest straight-line distance (diagonal) between two opposite corners of the room so that the Wi-Fi signal can reach the farthest corner efficiently. Write a Java program to calculate the diagonal distance using Pythagoras' Theorem, you can assume the length and width of the room is 10 and 13 respectively.

Question: 5

Design a Java program to calculate Sin and Cos values from a right-angled triangle.



Assume the values of a and b are 4.5 and 9.5 respectively. Finally, print the Sin and Cos values of angle A and angle B (SinA, CosA, SinB, CosB). The formulas to calculate these values are given below.

Hint: You need the values for all 3 sides to calculate both sin and cos. You are given only a and b. How would you get the value of c? You'll need the help of Math.sqrt().

Trigonometry formulas			
$\sin(A) = \frac{a}{c}$	$\cos(A) = \frac{b}{c}$	$\sin(B) = \frac{b}{c}$	$\cos(B) = \frac{a}{c}$

LAB ASSIGNMENT # 02

Branching



CSE110 | Programming Language I

	LAB TASKS	HOME TASKS
CODING	05	06
TRACING	01	02

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LAB TASKS
[NO NEED TO SUBMIT]

Question: 1

Write the Java code of a program to find the largest among three numbers entered by the user.

Sample Input	Sample Output
Enter Three Numbers: 100 23 -4	Largest Number: 100
Enter Three Numbers: -7 -36 -5	Largest Number: -5

Question: 2

You are developing a student grading system for a university portal. One of the features of this system is to automatically determine a student's letter grade based on their numerical score, which is entered by the user. The grading policy is defined as follows:

Scores	90-100	85-89	70-84	57-69	50-56	<50
Grades	A	A-	B	C	D	F

Sample Input	Output
45	Your grade is F
87	Your grade is A-

Question: 3

Write a Java code of a program that takes an integer number as user input and then determines if that number is divisible by both 5 and 7; otherwise display “No”. For example, numbers like 35, 70, 105, 140, 175, 210, 245, 280 etc. can be divisible by both 5 and 7.

Sample Input	Output
15	Invalid: Divisible by 5 Only
28	Invalid: Divisible by 7 Only
105	Divisible by Both
36	No

Question: 4

The Registrar’s Office at a University is preparing the academic calendar for upcoming sessions. Before finalizing the schedule, they need to determine whether a given year will have February 29 because leap years affect semester dates, holidays, and exam schedules.

To automate this, they’ve asked you to write a simple Java program that checks whether a year is a leap year or not, following the standard rules:

- A year is a leap year if it is divisible by 4.
- However, if the year is a century (divisible by 100), it must also be divisible by 400 to be a leap year.

Sample Input	Output
2020	2020 is a leap year
2001	2001 is not a leap year
1900	1900 is not a leap year
2000	2000 is a leap year

Question: 5

It's 11:59 PM, and Harun is desperately trying to finish a math assignment before the deadline. The final question involves a strange piecewise function, and Harun needs to write a program to verify the answers before submitting.

The function is defined as follows:

$$f(x) = \begin{cases} 2x, & x < 0 \\ x + 1, & 0 \leq x < 2 \\ x^2 - 1, & 2 \leq x < 5 \\ 3x^2 + 2, & x \geq 5 \end{cases}$$

Your task is to help Harun by writing a Java program that takes the value of x as input from the user and displays the corresponding value of f(x) according to the formula.

Sample Input	Output
-3	output: -6
1	output: 2
4	output: 15
10	output: 302

Question: 6

What will be the output of the following program? Show the workings.

1	<code>public class Tracing1 {</code>	Output
2	<code> public static void main(String[] args) {</code>	
3	<code> int num1 = 10;</code>	
4	<code> int num2 = -3;</code>	
5	<code> int num3 = -1;</code>	
6	<code> int sum = num1 + num2 + num3;</code>	
7	<code> if (num3 < 0) {</code>	
8	<code> System.out.println(num3 * (-2));</code>	
9	<code> }</code>	
10	<code> else {</code>	
11	<code> System.out.println(sum);</code>	
12	<code> }</code>	
13	<code> if (num1 < 5) {</code>	
14	<code> System.out.println(num1 + 10);</code>	
15	<code> }</code>	
16	<code> else if (num2 == -3) {</code>	
17	<code> num2 = num1;</code>	
18	<code> System.out.println(num2);</code>	
19	<code> }</code>	
20	<code> else {</code>	
21	<code> System.out.println(num1 + num2 + num3);</code>	
22	<code> }</code>	
23	<code> if (num1 > 15) {</code>	
24	<code> System.out.println(num1);</code>	
25	<code> }</code>	
26	<code> if (num2 == 0) {</code>	
27	<code> System.out.println(num2 + num3);</code>	

28	}
29	else {
30	System.out.println(num3);
31	}
32	if (sum != 0) {
33	System.out.println(100);
34	}
35	else {
36	System.out.println(sum + 100);
37	}
38	if (num1 > 0 && num2 < 0) {
39	System.out.println(num1 == num2);
40	}
41	else {
42	System.out.println("End");
43	}
44	}
45	}

HOME TASKS

Question: 1

Write a Java code of a program that takes an integer number as user input and then determines if that number is divisible by either 5 or 7 but not both; otherwise display “No”. For example, numbers like 35, 70, 105, 140, 175, 210, 245, 280 etc. can be divisible by both 5 and 7. These numbers are invalid.

Sample Input	Output
15	Divisible by 5 Only
28	Divisible by 7 Only
105	Invalid: Divisible by both
36	No

Question: 2

A new Smart Electric Car Charging Station has been set up in your city.

The station automatically calculates the total cost for charging based on the amount of electricity consumed (in kWh) and the type of user.

To make the billing fair and environmentally conscious, the station follows these rules:

Base rate: 15 Taka per kWh.

- Discounts and surcharges:
 - Students (under 18 years old) get a 20% discount on the total bill.
 - Regular users (18 to 60 years old) pay the normal rate.
 - Senior citizens (above 60 years old) get a 10% discount.
- High usage surcharge:
 - If total energy consumption exceeds 100 kWh, a 5% surcharge is added after applying any discount.

Sample Input	Output
Age: 16 Electricity Consumed: 50	Final Bill: 600.0 Taka
Age: 65 Electricity Consumed: 120	Final Bill: 1701.0 Taka

Question: 3

Write a Java program that takes 3 float numbers as input from the user and prints the maximum and minimum number from the inputs.

Sample Input	Output
18.83 4.02 83.12	Maximum number is 83.12 Minimum number is 4.02
-26.45 -0.02 -13.56	Maximum number is -0.02 Minimum number is -26.45

Question: 4

A triangle has 3 sides. Write a program which asks the users for input. Based on the input, your program should output whether it is an Equilateral, Isosceles or Scalene.

- Equilateral triangle has three sides with equal length
- Isosceles triangle has two sides with equal length and another side is different
- Scalene triangle has different lengths in each side

Sample Input	Output
5 2 4	This is a Scalene triangle
5 5 3	This is a Isosceles triangle
3 3 3	This is a Equilateral triangle

Question: 5

Suppose you are hired by a supershop named Bastob. Now, your task is to create a Java program that will help the cashier calculate the change to be returned.

The program takes two inputs. The first input is an integer number which is the amount of money to be paid in taka and the second integer is the amount of money the customer gave to the cashier. Your program should print the following:

- If the customer gave more money than the actual amount, print change the cashier should return in notes and coins.
- If the customer gave less money than the amount to be paid, then print the amount the customer needs to pay.

Consider the following denomination for notes and coins in taka:

Notes: 100, 50, 20, 10.

Coins: 5, 2, 1.

Sample Input	Output
Enter the amount the customer need to pay(Taka) 35 Enter the amount, customer gave(Taka) 53	The returned amount is 18 taka. 100 taka note: 0 50 taka note: 0 20 taka note: 0 10 taka note: 1 5 taka coin: 1 2 taka coin: 1 1 taka coin: 1
Enter the amount the customer need to pay(Taka) 60 Enter the amount, customer gave(Taka) 500	The returned amount is 440 taka. 100 taka note: 4 50 taka note: 0 20 taka note: 2 10 taka note: 0 5 taka coin: 0 2 taka coin: 0 1 taka coin: 0
Enter the amount the customer need to pay(Taka) 50 Enter the amount, customer gave(Taka) 50	The returned amount is 0 taka.
Enter the amount the customer need to pay(Taka) 550 Enter the amount, customer gave(Taka) 520	Please pay 30 taka more.

Question: 6

Write a Java program that reads three numbers and prints "All numbers are equal" if all three numbers are equal, "All numbers are different" if all three numbers are different and "Neither all are equal nor different" otherwise.

User Input	Output
Input the 1st number: 2345 Input the 2nd number: 2452 Input the 3rd number: 4532	All numbers are different
Input the 1st number: 230 Input the 2nd number: 230 Input the 3rd number: 112	Neither all are equal or different

Question: 7

What will be the output of the following program? Your answer will not be accepted without the workings.

1	public class Tracing2 {
2	public static void main(String[] args) {
3	boolean var1, var2, var3, var4, var5, var6;
4	boolean result1, result2, result3, result4, result5;
5	boolean result6, result7, result8, result9, result10;
6	var1 = var2 = var3 = var4 = var5 = var6 = false;
7	result1 = result2 = result3 = result4 = result5 = false;
8	result6 = result7 = result8 = result9 = result10 = false;
9	var1 = (!false false) && true;
10	var2 = var1 && true;
11	var3 = false && !true;
12	var4 = true;
13	var5 = false;
14	var6 = var3 && true;
15	result1 = (var1 && var2) && (40 % 3 > 45) (var5 && var6);
16	result2 = (var1 var2) (result1 && false);
17	result3 = (var1 && result1) result2 var5;
18	result4 = (var1 var2) ((var3 && var1) && false);
19	result5 = (var1 && var2) && (result3 var1);
20	result6 = ((var3 !var2) && result5) true;
21	result7 = (var4 && result1) && ((result1 && false) true);
22	result8 = ((var1 && result3) && (!var5 var6)) && true;
23	result9 = ((result2 && var2) (!result7 && var1)) && !false;
24	result10 = !(var1 && true);
25	System.out.println(result1 + " " + result2);
26	System.out.println(result3 + " " + result4);
27	System.out.println(result5 + " " + result6);
28	System.out.println(result7 + " " + result8);
29	System.out.println(result9 + " " + result10);
30	}
31	}

Question: 8

What will be the output of the following program? Your answer will not be accepted without the workings.

1	<code>public class Tracing3 {</code>
2	<code> public static void main(String[] args) {</code>
3	<code> int p = 5;</code>
4	<code> int q = 6;</code>
5	<code> int r = 9;</code>
6	<code> int sum = 0;</code>
7	<code> if (p < 12) {</code>
8	<code> System.out.println(r + 2);</code>
9	<code> }</code>
10	<code> else {</code>
11	<code> System.out.println(r + p);</code>
12	<code> }</code>
13	<code> if (q > 20){</code>
14	<code> System.out.println(r + 19);</code>
15	<code> }</code>
16	<code> else if (q <= 6) {</code>
17	<code> System.out.println(q + 3);</code>
18	<code> }</code>
19	<code> else{</code>
20	<code> System.out.println(p + q + r);</code>
21	<code> }</code>
22	<code> if (r > 15) {</code>
23	<code> System.out.println(r);</code>
24	<code> }</code>
25	<code> else if (r == 0) {</code>
26	<code> System.out.println(p + q);</code>
27	<code> }</code>
28	<code> else {</code>
29	<code> System.out.println(p);</code>
30	<code> }</code>
31	<code> if (sum != 0) {</code>
32	<code> System.out.println(3);</code>
33	<code> }</code>
34	<code> else {</code>
35	<code> System.out.println(sum + 32);</code>
36	<code> }</code>
37	<code> if(p > 0 && r < 10){</code>

38	<code>System.out.println(p + r);</code>
39	<code>}</code>
40	<code>else {</code>
41	<code>System.out.println(p - r);</code>
42	<code>}</code>
43	<code>}</code>
44	<code>}</code>

LAB ASSIGNMENT # 03

Linear Loop



CSE110 | Programming Language I

	LAB TASKS	HOME TASKS
CODING	07	08
TRACING	01	02

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LAB TASKS
[NO NEED TO SUBMIT]

Question: 1

Write a Java code that would print the following sequences using while/for loop:

- I. **24, 18, 12, 6, 0, -6**
- II. **18, -27, 36, -45, 54, -63**

Question: 2

Write a Java program that will take N numbers from the user and find their sum and average using a for loop.

Sample Input:

N = 5

Input the 5 numbers:

1
2
3
4
5

Expected Output:

The sum of 5 no is: 15

The Average is: 3.0

Question: 3

You are the engineer maintaining the power distribution system of NeoVolt City, a futuristic city powered by solar grids.

Each day the power grid generates and loses energy based on environmental conditions:

- If the day number is divisible by 3, a solar storm occurs, power drops by 45 units
- If divisible by 5, wind turbines overproduce, power increases by 25 units
- Otherwise, normal operation, power increases by 10 units

The grid starts with 40 units of power.

If the power ever exceeds 500, excess power is wasted and reduced by 50 units to protect it from overloading.

If power falls below 0, a blackout occurs, stop the simulation immediately.

Sample Input	Sample Output
Enter Number of Days: 5	Final Power: 50

Enter Number of Days: 12	Blackout on Day 9
--------------------------	-------------------

Question: 4

Write a Java program that will keep taking integer numbers as inputs from the user and print the square of those numbers until it gets a negative number and then stop.

Sample Input/Output: (The purple numbers are input.)

Enter Number: 2

$2^2 = 4$

Enter Number: 6

$6^2 = 36$

Enter Number: 1

$1^2 = 1$

Enter Number: 4

$4^2 = 16$

Enter Number: -5

Question: 5

Write a Java program that will calculate the value of y where the expression of y is as follows :

$$y = 1^2 - 2^2 + 3^2 - 4^2 + 5^2 \dots \dots + n^2$$

Sample Input	Output	Explanation
5	15	$y = 1 - 4 + 9 - 16 + 25 = 15$
10	-55	$y = 1 - 4 + 9 - 16 + 25 - 36 + 49 - 64 + 81 - 100 = -55$
20	-210	$y = 1 - 4 + 9 - 16 + 25 \dots \dots - 400 = -210$

Question: 6

Write a program in Java that asks the user for an **integer** input and counts the number of digits in the number.

Hint: You may keep dividing the number by ten and count how many times this can be done until the number becomes 0.

Sample Input	Output
7546	Total digits = 4

Question: 7

In a sunny garden, a gardener waters plants for H hours to help them grow. He starts with W water units. Each hour, he uses 3 units per plant, but there are 5 plants. If the hour is hot (hour number is a multiple of 2), use 4 units per plant instead. Subtract water each hour. If the water goes below 0 in any hour, stop and say the hour. If all hours are done, say the remaining water.

Sample Input	Sample Output
Enter Hour (H): 5 Enter Water Units (W): 10	Stopped at Hour 1
Enter Hour (H): 5 Enter Water Units (W): 100	All Watered 15 Water Unit Left
Explanation: Plants=5 (fixed) Water=100 (Starting with 100 Water Units) Hour-1 (odd), so $5 * 3 = 15$, Remaining Water: 85 Hour-2 (even), so $5 * 4 = 20$, Remaining Water: 65 Hour-3 (odd), so $5 * 3 = 15$, Remaining Water: 50 Hour-4 (even), so $5 * 4 = 20$, Remaining Water: 30 Hour-5 (odd), so $5 * 3 = 15$, Remaining Water: 15	

Question: 8

1	public class Tracel{
2	public static void main(String args[]){
3	int x = 0, p = 0, sum = 0;
4	p = 1;
5	x = 2;
6	double q;
7	sum = 0;
8	while (p < 12){
9	q = x + p-(sum+7/3)/3.0%2 ;
10	sum = sum + (++x) + (int)q;
11	System.out.println(sum++);
12	if (x > 5){
13	p += 4/2;
14	}
15	else {
16	p += 3%1;
17	}
18	}
19	sum = sum + p;
20	System.out.println(sum) ;
21	}
22	}

HOME TASKS

Question: 1

Write a Java program that will take a positive integer n as input and print all the numbers from 0 to n which are **divisible by 5 but not divisible by 3**.

Sample Input

40

Sample Output

5
10
20
25
35
40

Question: 2

In Hogwarts, apprentice wizards brew magic potions using glowing herbs and sparkling crystals collected from the Whispering Woods. You have a stash of H herbs and C crystals. Each successful potion requires exactly 3 herbs and 2 crystals. Brew as many potions as possible without running short mid-mix stop when you can't complete another full potion. After brewing, check the total potions: if even, the batch is perfectly balanced ("Stable Elixir"); if odd, it's unpredictably fizzy ("Volatile Brew"). Output the number of potions and the balance message.

Sample Input	Sample Output
Please Enter number of Herbs (H): 25 Please Enter number of Crystals (C): 13	Potion-1 created Remaining Herbs: 22, Remaining Crystals: 11 Potion-2 created Remaining Herbs: 19, Remaining Crystals: 9 Potion-3 created Remaining Herbs: 16, Remaining Crystals: 7 Potion-4 created Remaining Herbs: 13, Remaining Crystals: 5 Potion-5 created Remaining Herbs: 10, Remaining Crystals: 3 Potion-6 created Remaining Herbs: 7, Remaining Crystals: 1 Potions Created: 6

	Stable Elixir
Please Enter number of Herbs (H): 9 Please Enter number of Crystals (C): 6	Potion-1 created Remaining Herbs: 6, Remaining Crystals: 4 Potion-2 created Remaining Herbs: 3, Remaining Crystals: 2 Potion-3 created Remaining Herbs: 0, Remaining Crystals: 0 Potions Created: 3 Volatile Elixir

Question: 3

In a sunny mountain cave, a baby dragon named Charmander plays a fun game to evolve. Charmander starts with E energy points. He must blow fire across N small rivers. For each river i, the distance is D units. If D is 5 or less, it costs 2 energy. If D is more than 5, it costs $D / 2$ energy (whole numbers only). Take away the energy after each blow. If energy goes below 0, Charmander gets tired and stops. Tell the river where he stops, or if he finishes all, say how much energy is left.

Sample Input	Sample Output
Energy (E): 15 Number of River (N): 3 Enter River Distance D1: 4 Enter River Distance D2: 6 Enter River Distance D3: 3	All Done 8 energy Left
Energy (E): 19 Number of River (N): 8 Enter River Distance D1: 14 Enter River Distance D2: 12 Enter River Distance D3: 9 Enter River Distance D4: 2 Enter River Distance D5: 15	Tired at River 5
Explanations: Energy is 19 and needs to blow fire across 8 rivers In first river distance (D) is 14 which is greater than 5 so it costs 7 (half of D) so remaining energy (19-7=12) In second river distance (D) is 12 which is greater than 5 so it costs 6 (half of D) so remaining energy (12-6=6) In third river distance (D) is 9 which is greater than 5 so it costs 4 (half of D) so remaining energy (6-4=2)	

In fourth river distance (D) is 2 which is less than 5 so it costs 2 so remaining energy (2-2=0)
In fifth river distance (D) is 15, but there is no energy left, therefore it got tired at River 5

Question: 4

Write a program in Java that prints the individual digits of an integer number (user input) backward (From right to left).

Sample Input	Output
32768	8, 6, 7, 2, 3

Hint: First to get the digit from the right side, we can take the remainder of the number using the modulus (%) operator i.e. mod 10 to get the rightmost digit and print it. For dropping the last digit, we can perform division by 10 on the number and then continue the same to print the other digits as shown below.

Explanation:

$$32768 \% 10 = 8$$

$$32768 / 10 = 3276$$

Then,

$$3276 \% 10 = 6$$

$$3276 / 10 = 327$$

and so on

$$327 \% 10 = 7$$

$$327 / 10 = 32$$

$$32 \% 10 = 2$$

$$32 / 10 = 3$$

$$3 \% 10 = 3$$

$$3 / 10 = 0$$

Done! When the number becomes 0 you can end your loop.

Question: 5

In a secret city, you are on a mission to open a high-tech vault that protects a treasure!
The vault has a N-digit secret code (like 32768). To open it, you must say each digit one by one, from left to right, exactly as shown on the screen. Your smart watch can read the full number, but you need to break it into digits and speak them out loud like:

"3... 2... 7... 6... 8..."

If you get even one digit wrong, the alarm goes off!

Write a Java program that:

- Asks you: "Enter the N-digit vault code:"
- Reads the number
- Prints the digits forward, separated by commas

Rules:

- You cannot use **Math.pow()**
- You must count the digits first
- Then use division and remainder to pull out each digit
- Print them from left to right

[Instructions From Headquarters:

Start by finding how many digits (N). For 32768 → 5 digits. Then start with 10,000 (that's $10^{(N-1)}$).

Divide the code by 10,000 → get 3

Take remainder → left with 2768

Now divide the divider by 10 → 1000

Repeat: divide, print, remainder, shrink divider

Until you print all digits!]

Question: 6

Write a Java program that will take an integer as input and -

- A. Find out if the number is a prime number or not.
- B. Find out if the number is a perfect number or not.

[Prime Number: If a number has only two divisors, (1 and itself), then it is a prime number. Else, then it is not a prime number.

Perfect Number: A number is said to be a perfect number if the sum of its divisors, including 1 but not the number itself is equal to that number.]

Sample Input

6

Sample Output

6 is not a prime number

6 is a perfect number

Question: 7

Write a Java program which adds all numbers that are multiples of either 7 or 9 but not both, up to 600. Use only a single loop in your program. *[Ans: 39814]*

Question: 8

Write a Java code that asks an integer as input from the user and takes that many integer inputs. Your task is to count how many numbers are non-negative and negative.

Sample Input: (The purple numbers are input.)

Enter an integer: 9

Enter number 1: -8

Enter number 2: 33

Enter number 3: -100

Enter number 4: 10

Enter number 5: 0

Enter number 6: 5

Enter number 7: 10

Enter number 8: -4

Enter number 9: 4

Sample Output:

6 Non-negative Numbers

3 Negative Numbers

Question: 9

1	<code>public class Trace2 {</code>
2	<code> public static void main(String[] args) {</code>
3	<code> int x = 0, p = 0, sum = 0;</code>
4	<code> p = 1;</code>
5	<code> x = 2;</code>
6	<code> double q = 0.0;</code>
7	<code> sum = 5;</code>
8	<code> while (p < 15) {</code>
9	<code> q = x + p - (sum + (int) (7 / 4)) / 3.0 % 2;</code>
10	<code> sum = sum + x + (int) q;</code>
11	<code> x += 1;</code>
12	<code> System.out.println(sum);</code>
13	<code> if (x > 5) {</code>
14	<code> p += (int) (5 / 2);</code>
15	<code> }</code>
16	<code> else {</code>
17	<code> p += 10 % 3;</code>
18	<code> }</code>
19	<code> }</code>
20	<code> sum = sum + p;</code>
21	<code> System.out.println(sum);</code>
22	<code> }</code>
23	<code>}</code>

Question: 10

1	<code>public class Trace3 {</code>
2	<code> public static void main(String[] args) {</code>
3	<code> int m = 17, n = 13, p = 1, sum = 30;</code>
4	<code> while(0 < p%10){</code>
5	<code> if(m % 10 == 0){</code>
6	<code> sum = sum * m % n + p / n ;</code>
7	<code> }</code>
8	<code> else{</code>
9	<code> if(m % 4 == 0){</code>
10	<code> sum += n + (--m) ;</code>
11	<code> }</code>
12	<code> else{</code>
13	<code> sum -= m--;</code>
14	<code> }</code>
15	<code> }</code>
16	<code> p+=1;</code>
17	<code> System.out.println(sum) ;</code>
18	<code> }</code>
19	<code> System.out.println(!(n%13 == 0) && !false p>10);</code>
20	<code>}</code>
21	<code>}</code>

LAB ASSIGNMENT # 04

Nested Loop



CSE110 | Programming Language I

	LAB TASKS	HOME TASKS
CODING	04	04
TRACING	01	02

NOTE: You need to submit only the Home Tasks. Submit all the Tracing tasks handwritten respectively to your Lab Instructors before the next lab. Submit all the Homework Coding Tasks in the Google Form shared on buX.

LAB TASKS
[NO NEED TO SUBMIT]

Question: 1

In a virtual arena, there are N warriors numbered from 1 to N. Every warrior fights every other warrior exactly once. A warrior wins if their strength (their number) is divisible by the opponent's strength. Your job is to write a java code that calculates the total number of duels won in the entire tournament.

Sample Input	Sample Output
5	2 beats 1 3 beats 1 4 beats 1 4 beats 2 5 beats 1 Total wins: 5
8	2 beats 1 3 beats 1 4 beats 1 4 beats 2 5 beats 1 6 beats 1 6 beats 2 6 beats 3 7 beats 1 8 beats 1 8 beats 2 8 beats 4 Total wins: 12

Question: 2

Students at Hero Academia must complete missions and earn points in order to pass a grade. The homeroom teacher of class 1A wants to identify the top performer and give them a badge of honour. He provides you with the number of students (S), the number of missions (M) these students completed and for each student then gives you how many points they collected on each of these missions.

Write the java code to calculate each student's average points and help the teacher find the top performer. [N.B: All students receive points for all (M number of) missions.]

Sample Input	Sample Output
Students (S): 3 Missions (M): 3 Scores of Student 1: 10 9 8 Scores of Student 2: 9 10 10 Scores of Student 3: 7 8 9	Average point of Student 1: 9.0 Average point of Student 2: 9.67 Average point of Student 3: 8.0 Student 2 receives the badge of honour!!

Question: 3

You are testing SecureVault's safe locks. The program should ask for how many pairs (N) of codes to test, then for each pair, take two positive numbers and find their GCD using the Euclidean method. If any number is 0 or negative, print 0.

*[Hint: (Euclidean method) Two positive integers a and b . Assume a is greater than b , then divide a by b and find the remainder r . This can be expressed as $a = \text{quotient} * b + r$.*

If $r = 0$, b is the GCD of a and b and the loop terminated.

If $r \neq 0$, replace a with b and b with r and repeat the previous step.]

Example:

Steps	a	b	$a \% b$	Actions
1	48	18	12	$a \leftarrow 18, b \leftarrow 12$
2	18	12	6	$a \leftarrow 12, b \leftarrow 6$

3	12	6	0	Stop→GCD = 6
---	----	---	---	--------------

Input	Output
Enter number of pairs: 2 Pair 1: Enter first number: 40 Enter second number: 50 Pair 2: Enter first number: 0 Enter second number: 32	Pair 1: GCD = 10 Pair 2: GCD = 0

Question: 4

At Nimbus Tech, the HR analytics team is studying the “Stress Level” of a particular employee over “W” number of weeks. Each week the employee takes on a task more difficult than the previous week and adds on to his workload. This each week’s workload up to week “W”, is subtracted to find his Total Stress level.

For week 1, the workload is 1.

For week 2, it’s the sum of workload of previous week and this week’s new project (1+2).

For week 3, it’s (1+2+3). And so,

For week W, it’s (1+2+3+...+W)

The “Stress Level” also written as STRL is the result of subtracting all the weeks’ workload.

$$\text{STRL} = -(1) - (1+2) - (1+2+3) - \dots - (1+2+3+\dots+W)$$

Write a java code for the HR team to find the STRL of the employee when given “W” or number of weeks.

Sample Input	Sample Output
Enter number of Weeks, W: 4	STRL= -20
<u>Explanation:</u> Since W = 4, $\text{STRL} = - (1) - (1+2) - (1+2+3) - (1+2+3+4) = - 1 - 3 - 6 - 10 = -20$	

Question: 5

1	public class T1{
2	public static void main(String args[]){
3	int x = 0, y = 0;
4	int sum = 0;
5	while (x < 4){
6	y = x - 3;
7	while (y < 3){
8	sum = (sum % 3) + x - y * 3 ;
9	System.out.println(sum);
10	y = y + 1;
11	}
12	if (x > 5){
13	x++;
14	}
15	else{
16	x += 2;
17	}
18	}
19	}
20	}

HOME TASKS

Question: 1

NarcissiTech is a luxury telecom company that sells "Vanity Phone Numbers", special numbers that customers love because they are mathematically elegant. One of their flagship products is "Armstrong Vanity Numbers", where the number itself equals the sum of its digits each raised to the power of the total number of digits.

You need to write a Java program that asks the user to input a starting number and an ending number. The program must find and display all Armstrong numbers within this range (inclusive).

[An Armstrong number is a number whose digits' values, each raised to the power of the total number of digits, add up to the number itself.]

For example:

- 371 has 3 digits
- So we calculate:
 $3^3 + 7^3 + 1^3 = 27 + 343 + 1 = 371$
- Since the sum equals the original number, 371 is an Armstrong number.]

Question: 2

At Codewarts University of Magic and Programming, every student is assigned an unique numeric ID when they enroll. Professor Dumbledore, the head of the Department of Logical Arts, believes that students whose ID numbers are powers of 2 possess extraordinary logical abilities. Such students are said to have "Lucky IDs."

Your task is to help the university identify which students have Lucky IDs. You will first take an user input of the number of test cases (N) representing how many students' IDs need to be checked. For each student, take input of their ID number and determine whether it is a power of two. Display "Lucky ID" if the student's ID is a power of two, otherwise display "Not Lucky".

Sample Input	Sample Output
Enter number of students to check: 5 Enter student ID: 1 Enter student ID: 2 Enter student ID: 3 Enter student ID: 8	Lucky ID Lucky ID Not Lucky Lucky ID Not Lucky

Enter student ID: 10	
Explanation: Student ID: 1 = 2^0 Student ID: 2 = 2^1 Student ID: 3 = Not a power of 2. Student ID: 8 = 2^3 Student ID: 10 = Not a power of 2.	

Question: 3

At *ShopSmart Mart*, the owner keeps track of daily sales of three products for several days. At the end of each day, the shop calculates the total sales. To comply with local regulations, the shop applies taxes on total sales:

- If the total sales for the day equal to or greater than 100, a 2% tax is added.
- If the total sales equal to or greater than 200, a 5% tax is applied instead.
- If the total sales equal to or greater than 500, a 10% tax is applied instead.

The owner wants your help to display the sales table for each day, showing the total sales including any applicable tax.

You need to write a Java program that asks the user to input the number of days (N). Then, for each day, input the sales of the 3 products. For each day, calculate the total sales, apply the correct tax based on the total, and display the total sales including tax.

Sample Input	Sample Output
Enter number of days: 4 Enter sales for Day 1: 30 40 50 Enter sales for Day 2: 40 50 60 Enter sales for Day 3: 100 80 50 Enter sales for Day 4: 200	Day 1: Total Sales with Tax: 122.4 Day 2: Total Sales with Tax: 153.0 Day 3: Total Sales with Tax: 241.5 Day 4: Total Sales with Tax: 583.0

150 180	
------------	--

Explanation:

Sales for Day 1: $30+40+50 = 120$

Total sales= $120 + (2\% \text{ of } 120) = 122.4$

Sales for Day 2: $40+50+60 = 150$

Total sales= $150 + (2\% \text{ of } 150) = 153.0$

Sales for Day 3: $100+80+50 = 230$

Total sales= $230 + (5\% \text{ of } 230) = 241.5$

Sales for Day 4: $200+150+180 = 530$

Total sales= $530 + (10\% \text{ of } 530) = 583.0$

Question: 4

GymFit Elite uses an AI fitness system to calculate effective calorie burn per exercise with advanced rules. Your task is to write a java programme to calculate the average of earned calories. For each member, input the number of exercises and their raw calories. Each day a member can do the highest 3 exercises. If the calorie is greater than 350 then add 50% bonus, add 10% penalty to all if every exercise is less than 200, and give 50 calories flat bonus per exercise if the raw average is greater than 400. Now calculate the total calories earned and print the average calories earned a day.

Input	Output
Number of Members: 2 Exercises for Member-1: 3 Exercise-1: 300 Exercise-2: 400 Exercise-3: 800 Exercises for Member-2: 4 Exercise-1: 150 Exercise-2: 100 Exercise-3: 80	Average calories earned per day for Member-1: 750.000 (Can't do more than 3 exercise) Average calories earned per day for Member-2: 99.000

Explanation:

For first member (1), number of exercise is 3

Calorie for 3 exercises are 300, 400 and 800

So, rawSum = $300+400+800 = 1500$

As calorie burn for exercise 2 and 3 is more than 350 so 50% bonus is added to them

For 400, 200 will be added and for 800, 400 will be added, so total = $300 + 200 + 400 + 800 + 400 = 2100$

Average of rawSum is $1500 / 3 = 500$

The average of rawSum is greater than 400, therefore 50 bonus will be added for all three exercises. Therefore total is $2100 + 50 + 50 + 50 = 2250$

Finally total average is: $2250 / 3 = 750$

For second member (2), number of exercise is 4 which is greater than 3 so member need to perform maximum 3 exercises

Calorie for 3 exercises are 150, 100 and 80

So, rawSum = $150+100+80 = 330$

So, rawAvg = $330 / 3 = 110$

As calorie burn for exercise 1, 2 and 3 is not more than 200, 10% penalty will be given

$110 - 10\% = 99$

The average of rawSum is not greater than 400, therefore 50 bonus will not

be added for all three exercises.

Question: 5

1	public class T3
2	{
3	public static void main(String args[])
4	{
5	int x = 0, y = 0;
6	int sum = 0;
7	while (x < 10){
8	y = x - 3;
9	y = 40;
10	while (y > 22){
11	if ((sum > 30) && (sum < 40)){
12	sum = sum + x * 2 ;
13	}
14	else if ((sum > 40) && (sum < 50)){
15	sum = sum + x * 3;
16	}
17	else {
18	sum = sum + 23;
19	}
20	System.out.println(sum);
21	y = y - 10;
22	}
23	x += 2;
24	}
25	}
26	}

Question: 6

1	public class Tracing{
2	public static void main(String[] args) {
3	int sum = 0, r = 0, i = 10, j = -5000;
4	while (i < 10000) {
5	r = (i%2) + 1;
6	j = i + 2;
7	while (j < i){
8	r = 2-4+6-8+10-12;
9	j++;
10	}
11	sum = i % r + (i * r) - j + 5;
12	System.out.println(sum);
13	j--;
14	if (i > 18) {
15	sum += (++r) + i/3 + r + 12;
16	break;
17	}
18	else{
19	i += 3;
20	}
21	}
22	System.out.println(sum);
23	}
24	}

UNGRADED TASKS
(For Practice)

Questions: 1

In the ancient city of Luminallis, the Grand Scrollkeeper guards the Scroll of Know-it-all, a relic that grants you access to all knowledge in existence. To get the scroll you must pass the Scrollkeeper's test to find the sum of "unusual numbers". Unusual numbers are those that are not divisible by any number smaller than themselves, other than 1.

The scrollkeeper hands you two numbers: X (A starting point) and Y (The number of unusual numbers you must find). Write a java code to find the answer to the scrollkeeper's question.

Sample Input	Sample Output
X: 4 Y: 5	Sum: 53
<u>Explanation:</u> <i>5 unusual numbers starting from 4 are: 5, 7, 11, 13, 17</i> <i>Their sum: $5 + 7 + 11 + 13 + 17 = 53$</i>	

Question: 2

Read an integer N that is the number of test cases that follow. Each test case contains two integers X and Y. Print one output line for each test case that the sum of Y odd numbers from X including it if is the case. For example:

For the input 4 5, the output must be 45, that is: $5 + 7 + 9 + 11 + 13$

For the input 7 4, the output must be 40, that is: $7 + 9 + 11 + 13$

Sample Input	Sample Output
2 4 3 11 2	21 24
<u>Explanation:</u> Here, the 2 means there are two test cases. For each test case you have to take two inputs (X, Y) and print the sum of Y odd numbers starting from X.	

Question: 3

Write a Java program to take a positive integer N (where $N > 0$) as user input and print the first N prime numbers starting from 2. Your code should check all the positive integers starting from 2 and determine whether they are prime or not until N prime numbers are found.

Sample Input 1:

5

Sample Output 1:

2

3

5

7

11

Sample Input 2:

7

Sample Output 2:

2

3

5

7

11

13

17

Question: 4

Read one integer P ($12 \leq P \leq 1000$). Find all integer triples (a, b, c) with $1 \leq a < b < c$ such that $a + b + c = P$ and $a*a + b*b = c*c$. Print each triple on its own line as three numbers separated by single spaces, then print the total number of such triples on the last line. If there are no solutions, print 0 only.

Sample Input	Sample Output
60	10 24 26 15 20 25 Total- 2
120	20 48 52 24 45 51 30 40 50 Total- 120

LAB ASSIGNMENT # 05

PATTERN GENERATION



CSE110 | Programming Language I

	LAB TASKS	HOME TASKS
CODING	08	07
TRACING	00	00

NOTE: You need to submit only the Home Tasks. Submit all the Homework Coding Tasks in the Google Form shared on buX.

LAB TASKS
[NO NEED TO SUBMIT]

Rules:

- You are not allowed to use any array or String
- One only line where the word “String” may appear is in public static void main(**String** [] args) {
- The word “char” must not be anywhere in your solution

Overall hint: Imagine all outputs as a matrix of space and digit. Then count spaces and digits to find out the trend of increment or decrement of number of digits/spaces. Utilize several IF statements to control when ‘digit’ will be printed, when ‘space’ will be printed and when ‘enter’ will be printed. Nested loops may be utilized to print multiple lines.

Question: 1

Number Line

Sample Input	Sample Output
6	123456
7	1234567

Question: 2

Rectangle

Sample Input	Sample Output
4 6	123456 123456 123456 123456
8 4	1234 1234 1234 1234 1234 1234 1234 1234

Question: 3

Triangle - Left Justified

Sample Input	Sample Output
4	1 12 123 1234
6	1 12 123 1234 12345 123456

Question: 4

Triangle - Right Justified

Sample Input	Sample Output
4	4 34 234 1234
6	6 56 456 3456 23456 123456

Question: 5

Triangle - Isosceles

Sample Input	Sample Output
3	1 123 12345
4	1 123 12345 1234567

Question: 6

Hollow Rectangle

Hint 1: Print the character space (' ') in the middle.

Hint 2: You can re-use your solution to **PROBLEM 2) Rectangle** and use **IF** condition to selectively print the first and last digit of each line and all digits of the first and last line.

Sample Input	Sample Output
4 5	12345 1 5 1 5 12345
5 4	1234 1 4 1 4 1 4 1234

Question: 7

Hollow Triangle - Left Justified

Sample Input	Sample Output
5	1 12 1 3 1 4 12345
6	1 12 1 3 1 4 1 5 123456

Question: 8

Palindrome

Sample Input	Sample Output
5	123454321
7	1234567654321

HOME TASKS

Question: 1

Rhombus

Sample Input	Sample Output
3	1 123 12345 123 1
4	1 123 12345 1234567 12345 123 1

Question: 2

Hollow Triangle - Right Justified

Sample Input	Sample Output
5	5 45 3 5 2 5 12345
6	6 56 4 6 3 6 2 6 123456

Question: 3

Hollow Triangle – Isosceles

Sample Input	Sample Output
3	1 1 3 12345
4	1 1 3 1 5 1234567

Question: 4

Hollow Rhombus

Sample Input	Sample Output
3	1 1 3 1 5 1 3 1
4	1 1 3 1 5 1 7 1 5 1 3 1

Question: 5

Palindromic Triangle

Sample Input	Sample Output
5	1 121 12321 1234321 123454321
4	1 121 12321 1234321

Question: 6

Downward Palindromic Triangle

Sample Input	Sample Output
5	123454321 1234321 12321 121 1
4	1234321 12321 121 1

Question: 7

Hourglass

Sample Input	Sample Output
5	12345 123 1 123 12345
7	1234567 12345 123 1 123 12345 1234567

UNGRADED TASKS

Question: 1

You're a master baker at Golden Crust Bakery. Every morning, you make n loaves. But the dough must rise in stages. For each loaf, you check every 5 minutes:

- If it's not risen enough, print . (still rising).
- If it just rose, print * (ready!).
- Each loaf rises at a different minute (based on its number).

Write a program to print the rising status of the first n loaves over 20 minutes.

Sample Input:
Enter number of Loaves: 3
Sample Output:
<pre>.....****</pre>

Question: 2

You're a domino artist. You line up n dominoes in a row.

- When the first falls, it knocks the next every 1 second.
- Print the falling wave over n seconds.

Sample Input	Sample Output
Enter Time: 5	<pre>/ // /// //// /////</pre>
Enter Time: 8	<pre>/ // /// //// ///// //////// ///////// /////////</pre>

Question: 3

You're controlling the stage lights for a K-pop concert. There are 8 lights. Every second, the pattern shifts left. Print the first 8 seconds of the show. Light on = *, off = .

Sample Output

```
* * * * . . . .
* * * . . . . *
* * . . . . * *
* . . . . * * *
. . . . * * * *
. . . * * * * .
. . * * * * .
. * * * * .
* * * * . . . .
```

LAB ASSIGNMENT # 07

Arrays



CSE110 | Programming Language I

	LAB TASKS	HOME TASKS
CODING	05	05
TRACING	01	03

NOTE: You need to submit only the Home Tasks. Submit all the Flowchart or Tracing tasks hand drawn or handwritten, respectively to your Lab Instructors before the next lab. Submit all the Homework Coding Tasks in the Google Form shared on buX.

LAB TASKS
[NO NEED TO SUBMIT]

Question: 1

Write a Java program that will take N integer numbers from the user and create an array of length N.

- A. Print the elements of the array with their indices.
- B. Take another integer input from the user, resize the array by length 1, and add the new integer value to the array. Print the resized array.

Sample Input	Sample Output
N = 5 Enter a number: 11 Enter a number: 22 Enter a number: 33 Enter a number: 44 Enter a number: 55 Enter another number: 101	The elements of the array are: 0: 11 1: 22 2: 33 3: 44 4: 55 After resizing the array: 11 22 33 44 55 101

Question: 2

Write a Java program that asks the user for the length of an array and then creates an integer array of that length by taking inputs from the user. Then,

- A. Reverse the array by creating a new array of the same length and print it. (Out-of Place)
- B. Reverse the array without creating any new arrays and print it. (In-Place)

Sample Input	Sample Output
Enter the length of the array: 5 Enter a number: 7 Enter a number: -31 Enter a number: 344 Enter a number: 97 Enter a number: 100	Reversed using a new array: 100 97 344 -31 7 Reversed the original array: 100 97 344 -31 7

Question: 3

You are given an integer array with duplicate values. Write a Java program to update the array by replacing the duplicate values of the array with zero. Then print the updated array. [Your code should work for any given integer array]

Given Array	Sample Output
<pre>int arr [] = {9, -5, 7, 9, -5, 5, 7};</pre>	<pre>Before removing duplicates: 9 -5 7 9 -5 5 7 After replacing duplicates with 0: 9 -5 7 0 0 5 0</pre>

Question: 4

You are given an integer array. You need to create a new array that will contain only the unique (distinct) elements of the given array. Finally, print the new array.

Given Array	Sample Output
<pre>int arr [] = {23,100,23,56,100};</pre>	<pre>Input array: 23 100 23 56 100 New array: 23 100 56</pre>
<pre>int arr [] = {-5,10,-7,-5};</pre>	<pre>Input array: -5 10 -7 -5 New array: -5 10 -7</pre>

Question: 5

Take an integer N input from the user and create an array of length N by taking the elements as user input. Then, print the array. Next, sort the array in descending order using Selection Sort technique. Lastly, print the sorted array.

Sample Input	Sample Output
N = 6 Enter a number: 3 Enter a number: 6 Enter a number: 1 Enter a number: 2 Enter a number: 8 Enter a number: 5	Original Array: 3 6 1 2 8 5 Sorted Array: 8 6 5 3 2 1

Question: 6

Trace the following code, create a tracing table, and write the outputs.

1	class TraceA{
2	public static void main(String args[]) {
3	int [] myArray = new int[5];
4	int [] b;
5	int idx1 = 0, idx2 = 0;
6	b = myArray;
7	while (idx1 < 5){
8	myArray[idx1] = idx1 + 11;
9	idx2 = 1;
10	while (idx2 < idx1){
11	myArray[idx1] = b[idx2 - 1] + myArray[idx2] - idx1;
12	idx2 = idx2 + 1;
13	}
14	System.out.println(myArray[idx2]);
15	idx1 = idx1 + 1;
16	}
17	}
18	}

HOME TASKS

Question: 1

Take an integer N input from the user and create an integer array of N numbers by taking inputs from the user. Then, print the array. Next, modify the array by changing the positive numbers by 1 and the negative numbers by 0. If the element is zero, then it will be unchanged. Lastly, print the modified array.

Sample Input	Sample Output
N = 4 3 4 -2 1	Original array: 3 4 -2 1 After modifying: 1 1 0 1
N = 3 -4 0 2	Original array: -4 0 2 After modifying: 0 0 1

Question: 2

Write a Java program that will take N integer numbers from the user and create an array of length N. Take another number from the user and print the index of the number where it is found first. If not found then print 'Element not found'.

Note: Think about how to apply the concept of flag and break in this task.

Sample Input 1 N = 7 Enter a number: 45 Enter a number: 0 Enter a number: 17 Enter a number: 11 Enter a number: -34 Enter a number: -100 Enter a number: 17 17	Sample Input 2 N = 5 Enter a number: 4 Enter a number: 99 Enter a number: 23 Enter a number: -67 Enter a number: 34 55
Sample Output 1 17 is at index 2	Sample Output 2 Element not found

Question: 3

Write a Java program that asks the user for the length of an array then creates a double data-type array of that length by taking inputs from the user.

Then do the following:

- a. Show the maximum element and its index from the array.
- b. Show the minimum element and its index from the array.
- c. Show the summation of all the elements from the array.
- d. Show the average of all the elements from the array.

Sample Input	Sample Output
Enter the length of the array: 5 Enter a number: 7.5 Enter a number: -31.2 Enter a number: 344.0 Enter a number: 97.1 Enter a number: 100.4	Maximum element 344.0 found at index 2 Minimum element -31.2 found at index 1 Summation: 517.8 Average: 103.56

Question: 4

Write a Java program that will take input of two arrays and elements from the user and check whether the second array is a subset of the first array. A subset is a set that contains only elements found in the original set.

Sample Input - 1	Sample Output - 1
Please enter the length of array 1: 5 Please enter the elements of the arr1: 5 3 2 72 8 Please enter the length of array 2: 3 Please enter the elements of the arr2: 5 3 72	Array 2 is a subset of Array 1.

Sample Input - 2	Sample Output - 2
Please enter the length of array 1: 5 Please enter the elements of the arr1: 7 2 33 1 6 Please enter the length of array 2: 3 Please enter the elements of the arr2:	Array 2 is not a subset of Array 1.

1	
8	
2	

Question: 5

You are given two arrays of the same length. The first array contains the marks of the students and the second array contains the name of the students. You need to sort the marks array in ascending order while maintaining the corresponding names of the students in the names array aligned with their respective marks. Use the Bubble sort technique to solve this problem. [Your code should work for any given arrays]

Given Array 1	Sample Output 1
<pre>int [] marks = {85, 90, 75, 44, 99}; String [] names = {"Bob", "Alice", "Max", "Marry", "Rosy"};</pre>	<pre>Sorted Array: 44 75 85 90 99 Marry Max Bob Alice Rosy</pre>

Given Array 2	Sample Output 2
<pre>int [] marks = {100, 47, 85, 94, 5, 50}; String [] names = {"Henry", "Mari", "Herry", "Jack", "Lily", "Oliver"};</pre>	<pre>Sorted Array: 5 47 50 85 94 100 Lily Mari Oliver Herry Jack Henry</pre>

Question: 6

Trace the following code, create a tracing table, and write the outputs.

1	class TraceB{
2	public static void main(String args[]){
3	int [] myArray = new int[10];
4	int index1 = 0, index2 =0;
5	while (index1 < 10){
6	myArray[index1] = index1 + 3;
7	index2 = 1;
8	while (index2 < index1){
9	myArray[index1] = myArray[index1] + myArray[index2] - index1;
10	index2 = index2 + 1;
11	}
12	System.out.println(myArray[index1]);
13	index1 = index1 + 1;
14	}
15	}
16	}

Question: 7

Trace the following code, create a tracing table, and write the outputs.

1	class TraceC {
2	public static void main(String args[]){
3	int [] myArray = new int[10];
4	int [] b;
5	int index1 = 0, index2 =0;
6	index1 = 0;
7	b = myArray;
8	while (index1 < 10){
9	myArray[index1] = index1 + 4;
10	index2 = 1;
11	while (index2 < index1){
12	myArray[index1] = b[index1] + myArray[index2] - index1;
13	index2 = index2 + 1;
14	}
15	System.out.println(myArray[index1]);
16	index1 = index1 + 1;
17	}
18	}
19	}

Question: 8

Trace the following code, create a tracing table, and write the outputs.

1	class TraceD{
2	public static void main(String args[]){
3	int[] arr1 = {3,2,0,1,5,6,7};
4	int[] arr2 = {30,20,40,11,55,-34,100};
5	int a1 = 0, a2 = 0;
6	while (a1<arr1.length-1){
7	arr2[a1] = arr1[a2]+ a1 - arr2[a2];
8	a2 = 1;
9	while (a2 < a1){
10	arr2[a1] = arr2[a1] + arr1[a2] - a2;
11	a2 = a2 + 1;
12	}
13	System.out.println(arr2[a1]);
14	a1 = a1 + 1;
15	}
16	System.out.println(arr2[arr1[a2]]);
17	}
18	}

LAB ASSIGNMENT # 08

Method



CSE110 | Programming Language I

	LAB TASKS	HOME TASKS
CODING	03	03
TRACING	01	02

NOTE: You need to submit only the Home Tasks. Submit all the Flowchart or Tracing tasks hand drawn or handwritten, respectively to your Lab Instructors before the next lab. Submit all the Homework Coding Tasks in the Google Form shared on buX.

Lab Tasks

(No Need to Submit)

Task 01

[A,B,C,D should be written in a single java file]

- A. Write a method called **evenChecker** that takes an **integer** number as its argument and prints whether the number is even or odd **inside the method**.

Sample Method Call	Sample Output
<code>evenChecker(10);</code>	Even!!
<code>evenChecker(17);</code>	Odd!!

- B. Write a method called **isEven** that takes an **integer** number as an argument and **returns** boolean true if the number is even otherwise **returns** boolean false.

Sample Method Call	Sample Output
<code>boolean result = isEven(10); System.out.println(result);</code>	true
<code>boolean result = isEven(17); System.out.println(result);</code>	false

- C. Write a method called **isPos** that takes an **integer** number as an argument and **returns** boolean true if the number is positive otherwise **returns** boolean false.

Sample Method Call	Sample Output
<code>boolean result = isPos(-5);</code>	false

System.out.println(result);	
boolean result = isPos(12); System.out.println(result);	true

D. Write a method called **sequence()** that takes an **integer** in its parameter called n. Now, if n is **positive** then it prints all the **even** numbers from **0 to n**, otherwise if n is **negative**, it prints all the **odd** numbers from **n to -1**.

Note: **You must call** the methods from **1B** and **1C**, otherwise this task would be **considered invalid**.

Sample Method Call	Sample Output	Explanation
sequence(10);	0 2 4 6 8 10	Here, 10 is positive so 0,2,4,6,8,10 were printed.
sequence(-7);	-7 -5 -3 -1	Here, -7 is negative so -7,-5,-3,-1 were printed.
sequence(7);	0 2 4 6	Here, 7 is positive so 0,2,4,6 were printed
sequence(-8);	-7 -5 -3 -1	Here, -8 is negative so -7,-5,-3,-1 were printed.

Task02

[A,B,C should be written in a single java file]

A. Write a method called **circleArea** that takes an **integer** radius in its parameter and **returns** the **area** of the circle.

Note: area of a circle is πr^2

Sample Method Call	Sample Output
double area = circleArea(5); System.out.println(area);	78.5398

- B. Write a method called **sphereVolume** that takes a **double** radius in its parameter and **returns** the **volume** of the sphere.

Note: volume of a sphere is $\frac{4}{3}\pi r^3$

Sample Method Call	Sample Output
<code>double volume = sphereVolume(5.0); System.out.println(volume);</code>	523.5987

- C. Write a method called **findSpace** that takes two values in its parameters one is an **integer** diameter and another one is a String. Using the given diameter, this method should calculate the Area of a circle or the Volume of a sphere depending on the value of the second parameter. Finally, it should print the result **inside the method**.

Note: **You must call** the method written in task **2A & 2B**, otherwise this task would be **considered invalid**.

Sample Method Call	Sample Output
<code>findSpace(10,"circle");</code>	78.5398
<code>findSpace(5,"sphere");</code>	65.4498
<code>findSpace(10,"square");</code>	Wrong Parameter

Task03

[A,B should be written in a single java file]

- A. Write a method called **isTriangle** that takes 3 integer numbers as arguments. The method will **return** the boolean true if the 3 sides can form a valid triangle otherwise it'll **return** the boolean false.

Note: In a valid triangle, the sum of **any** two sides will be greater than the third side.

Sample Method Call	Sample Output	Explanation
<code>boolean res = isTriangle(7,5,10); System.out.println(res);</code>	true	Here, $7+5>10$, $5+10>7$ also, $10+7>5$. Thus, these 3 sides can form a valid triangle.
<code>boolean res = isTriangle(3,2,1); System.out.println(res);</code>	false	Here, $1+2\leq 3$, thus, these 3 sides can NOT form a valid triangle.

B. Write a method called **triArea** that takes 3 sides of a triangle as 3 **integer** arguments. The method should calculate and print the area of the triangle only if it's a valid triangle otherwise print that it's not a valid triangle.

Area of triangle = $\sqrt{s(s-a)(s-b)(s-c)}$, where 's' is the semi-perimeter of the triangle. So, semi-perimeter = double $s = \text{perimeter}/2 = (a + b + c)/2$.

Note: **You must call** the method written in task **3A**, otherwise this task would be **considered invalid**.

Sample Method Call	Sample Output	Explanation
<code>triArea(3,2,1);</code>	Can't form triangle	Here, $1+2\leq 3$, thus, these 3 sides can NOT form a valid triangle.
<code>triArea(7,5,10);</code>	16.248	Here, 7,5,10 is able to form a valid triangle so, using the formula we get the area as 16.248

Task04

Trace the following code to generate the outputs. Show the necessary trace table.

1	public class Trace1{
2	public static int f3(int a, int c){
3	System.out.println("F3 begins");
4	System.out.println(a+c);
5	System.out.println("F3 ends soon");
6	return a*5;
7	}
8	public static String f1(int n){
9	System.out.println("F1 begins");
10	f2(5,n);
11	System.out.println("F1 ends soon");
12	return n+1+" from F1";
13	}
14	public static void main(String[] args){
15	System.out.println("Starting soon...");
16	int c = 99;
17	String s = f1(c);
18	System.out.println(s);
19	System.out.println("The End");
20	}
21	public static void f2(int c, int d){
22	System.out.println("F2 begins");
23	System.out.println(f3(c+1,d-1));
24	System.out.println("F2 ends");
25	}
26	}

Home Tasks

Task01

[A,B,C should be written in a single java file]

- A. Write a method called **isPrime** which takes an integer in its parameter to check whether a number is prime or not. If the number is prime then the method returns boolean **true** otherwise it returns boolean **false**.

Sample Input	Sample Output
<code>boolean check = isPrime(7); System.out.println(check);</code>	true
<code>boolean check = isPrime(15); System.out.println(check);</code>	false

- B. Write a method called **isPerfect** which takes an integer in its parameter to check whether a number is perfect or not. If the number is perfect then the method returns boolean **true** otherwise it returns boolean **false**.

Sample Input	Sample Output
<code>boolean check = isPerfect(6); System.out.println(check);</code>	true
<code>boolean check = isPerfect(33); System.out.println(check);</code>	false

- C. Write a method called **special_sum** that calculates the sum of all numbers that are either prime numbers or perfect up till the integer value given in its parameter. This integer value must be taken as user input and passed into the method.

Note: **You must call** the methods written in task 1A & 1B, otherwise this task will be **considered invalid**.

Sample Input	Sample Output	Output
8	<pre>int result = special_sum(8); System.out.println(result);</pre>	23
Explanation:	Between 1 to 8 the Prime numbers are 2,3,5,7 and 6 is a Perfect number. So, the summation is 2+3+5+7+6=23.	

Task02

[A,B,C should be written in a single java file]

- A.** Write a simple method called **showDots** that takes a number as an argument and then prints that amount of dots inside the method.

Note: You can use `System.out.print()` to avoid the next output being printed on the next line.

Sample Method Call	Sample Output
<code>showDots(5);</code>	
<code>showDots(3);</code>	...

- B.** Write a method called **show_palindrome** that takes a number as an argument and then prints a palindrome inside the method.

Note: You can use `System.out.print()` to avoid the next output being printed on the next line

Sample Method Call	Sample Output
<code>show_palindrome(5)</code>	123454321
<code>show_palindrome(3)</code>	12321

C. Write a method called **showDiamond** that takes an integer number as an argument and then prints a **palindromic diamond shape**. Moreover, the empty spaces surrounding the diamonds are filled with dots(.) .

Note: **You must call** the methods written in task **2A & 2B**, otherwise this task would be **considered invalid**.

Sample Method Call	Sample Output
showDiamond(5)	<pre>1..... ...121... ..12321.. .1234321. 123454321 .1234321. ..12321.. ...121...1..... </pre>
showDiamond(3)	<pre> ..1.. .121. 12321 .121. ..1.. </pre>

Task03

[A,B should be written in a single java file]

A. Write a method called **calcTax** that takes 2 arguments which are **your age** then **your salary**. The method must calculate and **return** the tax as per the following conditions:

- No tax if you are less than 18 years old.
- No tax if you get paid less than 10,000
- 7% tax if you get paid between 10,000 and 20,000 (both inclusive)
- 14% tax if you get paid more than 20,000

Sample Method Call	Output	Explanation
<code>double t = calcTax(16,20000); System.out.println(t);</code>	0.0	Here, the age is less than 18 so 0 tax.
<code>double t = calcTax(20,18000); System.out.println(t);</code>	1260.0	Here, the age is greater than 18 and income is between 10K-20K so tax is 7% of 18000 = 1260.

B. Write a method called **calcYearlyTax** that takes no arguments. Inside the method it should take **first input as your age** and then **12 other inputs** as income of each month of the year. The method must calculate and print Tax for each month and finally print the total Tax of the whole year based on the **6A conditions**.

Note: **You must call** the method written in task **3A**, otherwise this task would be **considered invalid**.

Sample Method Call	Sample Input	Sample Output
<code>calcYearlyTax()</code>	22 8000 15000	Month1 tax: 0 Month2 tax: 1050.0 Month3 tax: 3080.0

	22000	Month4 tax: 0
	2300	Month5 tax: 1071.0
	15300	Month6 tax: 2940.0
	21000	Month7 tax: 4760.0
	34000	Month8 tax: 0
	9000	Month9 tax: 3780.0
	27000	Month10 tax: 12320.0
	88000	Month11 tax: 4480.0
	32000	Month12 tax: 0
	7300	Total Yearly Tax: 33481.0

Task04

Trace the following code to generate the outputs. Show the necessary trace table.

1	public class Trace2{
2	public static void main (String[] args){
3	int a = 4, b = 7;
4	System.out.println(methodA(a,b));
5	}
6	public static double methodA(int m, int n){
7	int p = m+n-23, s = 0;
8	if (p<0){
9	System.out.println(p);
10	System.out.println(methodB(p+10));
11	s = methodB(p-10);
12	}
13	System.out.println(--s);
14	return p*m+s;
15	}
16	public static int methodB(int r){
17	int q = 6;
18	System.out.println(++r + q);
19	return q-r;
20	}
21	}

Task05

Trace the following code to generate the outputs. Show the necessary trace table.

1	public class Trace3{
2	public static boolean met1(int n1, int n2){
3	System.out.println("Method 1");
4	int n = n1+n2;
5	System.out.println(n);
6	return met3(n, n2)>n1;
7	}
8	public static int met2(int n, String s){
9	int p = 5;
10	System.out.println("Method 2");
11	System.out.println(met1(n,p));
12	return s.length();
13	}
14	public static double met3(int n, int p){
15	System.out.println("Method 3");
16	System.out.println(n/p);
17	return p;
18	}
19	public static void main (String[] args){
20	System.out.println("Main Method");
21	System.out.println(met2(6,"ABC"));
22	}
23	}

LAB ASSIGNMENT # 09

Recursive Method



CSE110 | Programming Language I

	LAB TASKS	HOME TASKS
CODING	04	04
TRACING	01	01

NOTE: You need to submit only the Home Tasks. Submit all the Flowchart or Tracing tasks hand drawn or handwritten, respectively to your Lab Instructors before the next lab. Submit all the Homework Coding Tasks in the Google Form shared on buX.

Lab Tasks

(No Need to Submit)

Task01

[A,B,C should be written in a single java file]

A. Write a method called **oneToN** that prints 1 till N recursively.

Hint: N is a number taken as input from the user and you need to print the numbers starting from 1 to N recursively.

Sample Input	Sample Method Call	Output
N = 5	oneToN(1,N);	1 2 3 4 5
N = 11	oneToN(1,N);	1 2 3 4 5 6 7 8 9 10 11

B. Write a method **nToOne** that prints from N to 1 recursively.

Hint: N is a number taken as input from the user and you need to print the numbers starting from N to 1.

Sample Input	Sample Method Call	Output
N = 6	nToOne(1,N);	6 5 4 3 2 1
N = 3	nToOne(1,N);	3 2 1

C. Write a method called **recursiveSum** to sum till N recursively.

Hint: N is a number taken as input from the user and you need to add the numbers starting from 1 to N recursively and print the sum.

Sample Input	Sample Method Call	Output
N = 4	System.out.println(recursiveSum(1,N));	10
N = 12	System.out.println(recursiveSum(1,N));	78

Task02

Write a **recursive method** called **reverseDigits** that takes an integer n as an argument and prints the digits of n in reverse order.

Hint: Think about how you solved it using loop

Sample Input	Sample Method Call	Output
n = 12345	reverseDigits(n);	5 4 3 2 1
n = 649	reverseDigits(n);	9 4 6
n = 1000	reverseDigits(n);	0 0 0 1

Task03

Write a **recursive method** called **sumDigits** that takes an integer n as an argument and sums up the digits of n then **returns** the result.

Hint: Think about how you would solve it using loop

Sample Input	Sample Method Call	Output
n = 12345	int x = sumDigits(n); System.out.println(x);	15
n = 649	int x = sumDigits(n); System.out.println(x);	19

Task04

Write a **recursive method** called **reverse_string(s, idx)** that returns the reverse of a given string **s**.

Sample Method Call	Output
System.out.println(reverse_string("Hello", 0))	olleH
System.out.println(reverse_string("swan", 0))	naws

Task05

Trace the following code to generate the outputs. Show the necessary trace table.

1	<code>public class ClassWork1{</code>
2	<code> public static int calculate(int n) {</code>
3	<code> if (n <= 0){</code>
4	<code> return 4;</code>
5	<code> }</code>
6	<code> else if (n % 2 != 0){</code>
7	<code> return n + calculate(n - 1);</code>
8	<code> }</code>
9	<code> else{</code>
10	<code> return n * calculate(n - 2);</code>
11	<code> }</code>
12	<code> }</code>
13	<code> public static void main(String[] args) {</code>
14	<code> int result = calculate(8);</code>
15	<code> System.out.println(result);</code>
16	<code> int result2 = calculate(5);</code>
17	<code> System.out.println(result2);</code>
18	<code> }</code>
19	<code>}</code>

Home Tasks

Task01

Write a **recursive method** called **factorial(n)** that returns the factorial of a number **n**. Assume $n \geq 0$.

Sample Input	Sample Method Call	Output
n = 5	int x = factorial(n); System.out.println(x);	120
n = 7	int x = factorial(n); System.out.println(x);	5040

Task02

Write a **recursive method** called **power(base, exponent)** that calculates base raised to the power of exponent (assume exponent is a non-negative integer).

Sample Method Call	Output
int x = power(5,3); System.out.println(x);	125
int x = power(8,4); System.out.println(x);	4096

Task03

Write a **recursive method** called **print_elements(arr, index)** that prints elements of an array starting from index to the end.

Given Array and Input	Sample Method Call	Output
<pre>int[] arr = {5,6,2,1,8,7}; int index = 2;</pre>	<pre>print_element(arr, index);</pre>	2 1 8 7
<pre>int[] arr = {13,12,19,21,31,55}; int index = 0;</pre>	<pre>print_element(arr, index);</pre>	13 12 19 21 31 55

Task04

The Fibonacci sequence is a series of numbers that starts with 0 & 1 and the rest of the numbers are generated by adding the immediate two numbers before it. It goes like this: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144 and so on.

In short, $\text{fibonacci}(0) = 0$, $\text{fibonacci}(1) = 1$ and $\text{fibonacci}(n) = \text{fibonacci}(n-1) + \text{fibonacci}(n-2)$.

Sample Method Call	Output
<pre>System.out.println(fibonacci(0));</pre>	0
<pre>System.out.println(fibonacci(1));</pre>	1
<pre>System.out.println(fibonacci(5));</pre>	5
<pre>System.out.println(fibonacci(9));</pre>	34

Task05

Trace the following code to generate the outputs. Show the necessary trace table.

1	<code>public class ClassWork2{</code>
2	<code> public static String fun(String s, int n){</code>
3	<code> if(s.length()==4){</code>
4	<code> return n+s+n;</code>
5	<code> } else if(n%2==0){</code>
6	<code> System.out.println(s+n+n+3);</code>
7	<code> return fun(s+n, n+3);</code>
8	<code> } else {</code>
9	<code> System.out.println(s+n+(n-1));</code>
10	<code> return fun(s+n, n-1);</code>
11	<code> }</code>
12	<code> }</code>
13	<code> public static void main(String[] args){</code>
14	<code> String s = fun("",1);</code>
15	<code> System.out.println(s);</code>
16	<code> }</code>
17	<code>}</code>